

An introduction to data and data analytics

The data practitioner's guide

Attributes of a data professional

Data professionals have many attributes that make them **effective** in **working with data**, **extracting insights**, and **deriving value** from data-driven approaches.



Analytical thinking

Tend to possess strong analytical skills and are capable of **breaking down complex problems into manageable components**. Can **identify patterns, trends, and insights** within data and use **critical thinking** to derive meaningful conclusions.



Technical proficiency

Tend to have a solid **understanding of data-related technologies, programming languages, and tools**. Proficient in data manipulation, analysis, and visualisation.



Curiosity and continuous learning

Tend to have a **natural curiosity** and a **passion** for **exploring** and **understanding data**. Eager to learn new techniques, tools, and methodologies to enhance their data analysis skills. Stay updated with the latest trends and advancements.

Attributes of a data professional



Attention to detail

Pays attention to detail when working with data. They ensure **data accuracy, completeness, and quality**. They are **meticulous** in data cleaning, pre-processing, and validation to minimise errors and biases that could impact analysis outcomes.



Collaboration and teamwork

Collaborate effectively with colleagues from different backgrounds, including stakeholders. They can **work together** to identify project goals, share insights, and contribute to data-driven decision-making.



Communication and storytelling

Skilled communicators who can effectively **convey complex data concepts**. Can translate data insights into clear, actionable recommendations and present findings through **effective use of visualisations** and **storytelling**.

While the specific combination and emphasis of attributes may vary depending on the role and organisation, these attributes are **commonly sought after** in data professionals, such as data analysts, data scientists, and data engineers.

Since these **attributes are in high demand**, it is valuable for data professionals to **develop these attributes**.

Key data skills

Data has become an integral part of daily life and many of these skills are **critical** but **transferable** in many careers – we are all **data users**.

As data professionals, there are various skills we should have.

Depending on the specific role and responsibilities, **additional skills** such as machine learning, domain knowledge, database systems, and programming may be required.



Key data tools

Data professionals use several **different tools** to leverage their data skills and knowledge, depending on their role. Although we don't need to be proficient in all of them, it's important to know what the toolkit of a data professional could look like.



Spreadsheets:

Google Sheets, Microsoft Excel, Numbers, LibreOffice Calc



Database management systems:

MySQL, PostgreSQL, Oracle, Microsoft SQL Server, MongoDB, SQLite, Apache Cassandra



Data visualisation and dashboarding tools:

Microsoft Power BI, Tableau, QlikView



Programming languages:

SQL, Python, R



Integrated development environments (IDEs):

Visual Studio Code, Jupyter Notebook, PyCharm, RStudio



Version control systems:

Git, Mercurial, Subversion



Cloud computing platforms:

Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure



Machine learning frameworks:

scikit-learn (Python), TensorFlow, Keras

And many more.

Problem solving

Problem solving is a critical skill for all data users because it provides **structured**, **transferable**, and **logical** ways of approaching data problems.

Depending on the problem we need to solve, we use critical thinking “tools” such as **logic trees** and **flowcharts** to understand the problem and plan a solution.

When we understand the problem, we apply a **variety of skills and tools to solve** the problem. The tools we could use include:



Spreadsheets



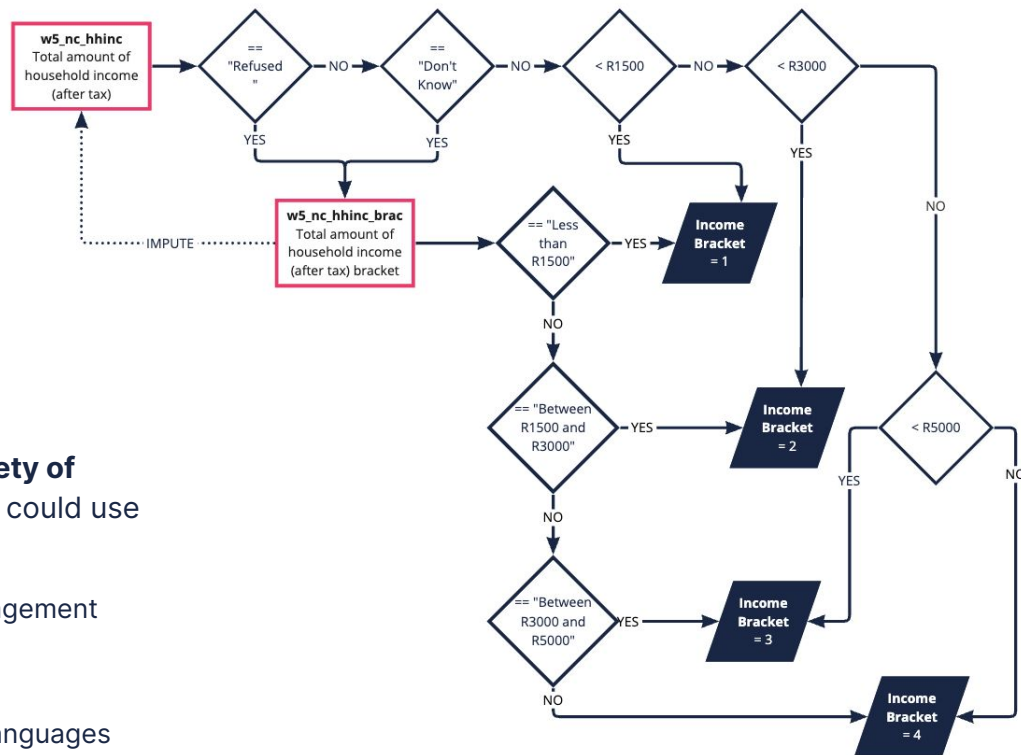
Database management systems



Data visualisation and dashboarding tools



Programming languages



Data preparation and analysis

Data preparation (cleaning) and **analysis** are critical skills for data professionals as they **enable effective decision-making, uncovering insights, identifying patterns, and detecting trends.**

The information extracted with these skills can **drive business strategy, optimisation, problem solving, and innovation.**

The tools we can use:



Spreadsheets



Database management systems

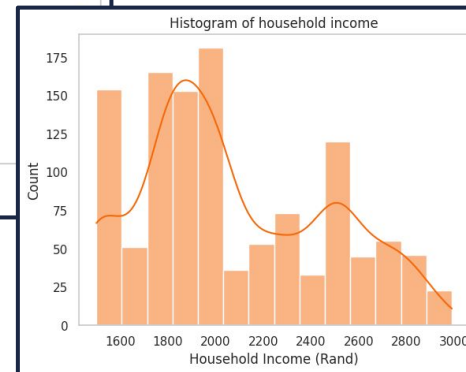
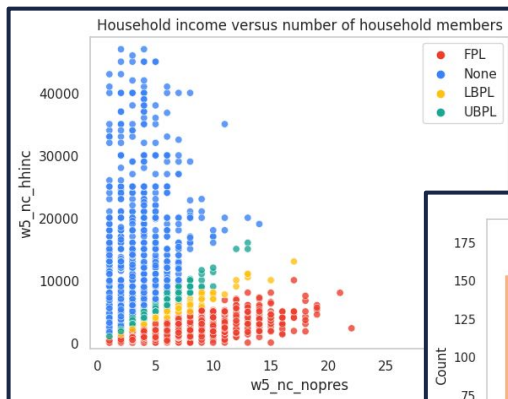


Data visualisation and dashboarding tools



Programming languages

Column	w5_nc_hhinc_brac	w5_nc_hhinc
Count non NaN	1561	5862
Count after drop "Refused" and "Don't Know" in w5_nc_hhinc_brac	1167	5468



Mathematics and statistics

Although **mathematics** and **statistics** may feel overwhelming to many, understanding the **fundamentals** of these fields makes understanding and **using data easier**.

Mathematics and statistics are **critical for data analysis, problem solving, and machine learning**.

We can apply this knowledge and skills across various tools, including:



Spreadsheets



Database management systems



Machine learning frameworks

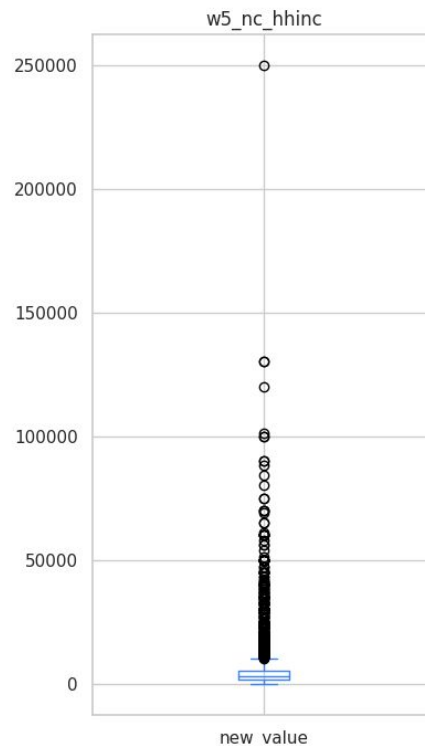


Programming languages

Column	w5_nc_hhinc
Count	4301.00
Mean	5653.48
Standard deviation	10057.09
Minimum	0.00
25%	1600.00
50%	3000.00
75%	5000.00
Maximum	250000.00

IQR = 3400
Higher outliers > R 10100

The number of outliers: 508 households (11.81%)



Communication and storytelling

Strong communication skills are essential for data practitioners to effectively communicate insights and findings to non-technical stakeholders.

The ability to **translate complex data concepts** into clear meaningful narratives helps organisations to make data-driven decisions.

We use **data visualisation, dashboards, reports, and presentations** to communicate data insights.

The tools we can use:



Spreadsheets



Data visualisation and dashboarding tools



Programming languages

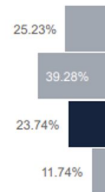
Unemployment and poverty in South Africa

INCOME PER CAPITA

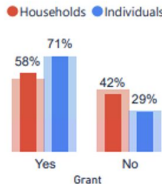


46.68

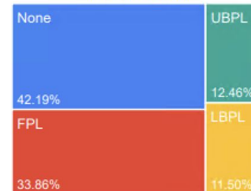
Mean Age 2017



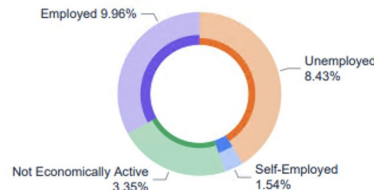
GRANTS AND EMPLOYED



POVERTY LEVEL



EMPLOYMENT



LFPR



UNEMPLOYMENT RATE

